

CHAPTER 1

A First Theory of Plausible Inference and Its Use in Continuous Domain Planning

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1. Introduction

AI systems often encounter difficulties when they are applied to real-world situations. This is largely because implemented systems typically are designed to work in micro-worlds that are, at best, approximations to some fragment of the real world. In writing domain knowledge of the micro-world, the implementor necessarily ignores most real-world relations and features and often simplifies even those that are represented. The apparent necessity of this simplification can be seen as an aspect of the *qualification* problem (McCarthy, 1969) which is well presented in (Genesereth, 1987) as:

Most universally quantified sentences will have to include an infinite number of qualifications if they are to be interpreted as accurate statements about the [real] world.

Since few implementors or computer systems are prepared to employ an infinite number of conditions on each piece of knowledge, one might believe that micro-worlds are an inescapable reality of AI. In fact, the above statement and the original conceptualization of the problem are in terms of situation calculus representations and perhaps apply only to situation calculus. However, Shoham (Shoham, 1986b) has argued, and it is generally accepted, that the qualification problem and other related aspects of the frame problem are “formalism independent” applying equally to all monotonic formal systems. In this paper we take issue with the last statement. We agree that the qualification problem transcends situation calculus, applying to other popular monotonic AI formalisms such as STRIPS-like operators (Chapman, 1987; Fikes, 1971; Pednault, 1988). However, we interpret this observation instead as an indictment of these formalisms. Non-monotonic logic (McCarthy, 1986; McDermott, 1980; McDermott, 1982; Reiter, 1980; Shoham, 1986a) offers an alternative but at a high computational cost. In addition, there are questions as to its adequacy (Hanks, 1986; McDermott, 1987). We propose a monotonic-like formalism that is quite different than existing monotonic and non-monotonic logics. While it is still the subject of on-going research, its theoretical and computational properties seem a promising fit to reasoning about action and change in the real world.

How can an AI system avoid believing unquestioningly in an idealized *a priori* micro-world? What would such a system look like? At the very least it must avoid brittleness with respect to real-world observations.

But what does mean to avoid *brittleness*? We will take it to mean that such a system must tolerate and adapt to any *actual* observation of the real world. In a micro-world system, any real-world observation may fundamentally contradict beliefs entailed by the system's axioms. Such contradictions lead to catastrophic failures of the inferential system that limit the applicability of such systems. Aside from cheating (imposing additional carefully chosen constraints on the world or artificially focusing the system's attention), there are two current approaches to fulfill the non-brittleness requirement. First, the system may be given the ability to alter its axiom set into one that entails (or at least does not contradict) the new observation. Work on theory revision (Ginsberg, 1991; Ourston, 1990; Pazzani, 1987; Wilkins, 1988) and some work on automated scientific discovery (Nordhausen, 1987; Rajamoney, 1988) is in this vein. The second approach is to employ general axioms with some built-in flexibility that can be adapted to embrace new observations as they arise. The various non-monotonic approaches to reasoning (Genesereth, 1987; McCarthy, 1986; Reiter, 1980) can be viewed this way.

The work reported in this paper is more in the spirit of the second. However, the axioms are weakened further than can be allowed through the introduction of *abnormal* relations or default modal operators. Indeed, it is questionable whether the system can be said to "believe" its initial axioms; acquired theorems have a great deal more force than do their supporting axioms. We first present the system that motivated this line of research, which learns to plan in continuous domains. After discussing its behavior we examine some of the system's interesting properties, particularly its knowledge-level learning behavior. We trace these features back to a level more basic than the fact that it works in continuous domains or the qualitative representations it employs or the fact that it is a planning system. We then introduce the beginnings of a new inferential system that we call *influential logic*.

2. Planning in Continuous Domains

From STRIPS and HACKER (Fikes, 1972; Sussman, 1973) through PRODIGY/EBL (Minton, 1988), learning has had a long, if not dominating, association with planning. Domain-independent planning, as commonly defined, has been shown to be intractable, or worse unde-

cidable (Chapman, 1987). We believe that learning holds the brightest, perhaps only, hope to salvage domain-independent planning. In particular, explanation-based learning (Mitchell, 1986; DeJong, 1986) may be extended to yield a kind of *adaptable* planning, which is achieved by explaining and integrating observed real-world changes into the system's beliefs. In this way we hope to avoid the quagmire of reactivity. Furthermore, the approach appears to shed light on largely-ignored issues such as computing effects of overlapping actions and dealing with continuous changes. These are the goals of our motivating system.

Consider the planning problem in the following scenario, in which an airplane is on the correct glide slope to land but with an airspeed that is 10 knots too fast.

The pilot gently begins to close the throttle. A moment later he eases back on the control yoke (the pilot's steering wheel) deflecting the elevator up and increases the plane's angle of attack. He stops closing the throttle and then holds the yoke steady once again. The plane slows while maintaining its current glide slope.

In this example, gentle and continuously coordinated changes in throttle and elevator controls are essential. The solution is problematic for current planning approaches since the required actions cannot easily be modeled as instantaneous. A temporal reasoning approach which allows change over time intervals may help a bit but its only contribution is to recognize that the two defined intervals (one for closing the throttle and one for moving the elevator) result in three different intervals: 1) throttle changing alone, 2) both throttle and elevator changing, and 3) elevator changing alone. This is an important realization but temporal reasoning does not ameliorate the cost of managing what *happens* during the middle interval. A system can model the middle interval only if there exists a single operator for the combined change. The domain theory implementor must have anticipated the need for this composite operator as well as operators for all other possible combination actions. This can result in an exponential increase in the number of operators, placing an excessive burden on both the implementor and the planning system.

Alternatively, we might try to recode the domain operators to disallow "large" changes. Instead, the **decrease-throttle** operator closes the throttle by some small amount Δ . Then the pilot's actions are

modeled as hundreds or perhaps thousands of appropriately interleaved instances of **decrease-throttle** and **pull-back-on-yoke** instead of just two overlapping large actions. Clearly, any desired degree of faithfulness to the real world can be achieved by choosing sufficiently small Δ 's. Unfortunately, this dramatically increases the length of the plan and the time to generate it since planning time is typically exponential in the length of the solution plan.

2.1 A Representation for Continuous Changes

Our primary interest lies in the use of *plausible* explanation-based learning. Plausibility allows explanations to be *conjectured* rather than entailed by the background knowledge. This shift away from absolute truth entailment takes pressure off of the system's inferential ability allowing non-brittle behavior. Plausible explanations are constructed from qualitative statements describing the domain which mediate between numerical values and more conventional symbolic representations. Plausible explanations are qualitative control analyses which are calibrated with the observed expert's behavior and refined if necessary when deficiencies are observed in the course of exercising the planning system.

Following qualitative reasoning terminology (Forbus, 1984) we call the continuous aspects of the world *quantities*. The term also arises informally with a similar meaning in control theory (Kuo, 1987). Examples of quantities are the position of the aircraft's throttle, the fuel flow through the intake manifold, the wing's angle of attack, and the speed of the plane. Quantities take on real values, they are continuous, and they may have limited extreme values. In general, many world quantities are changing simultaneously.

Some quantities can be directly manipulated by the planner such as the aircraft's throttle position or the setting of a radio's volume control. We call these quantities *controllable parametric quantities* (or *controllable parameters*). In control theory terminology these are *input variables*. As the controllable parameters are varied, values of other world quantities react in accordance with the laws of nature.

Quantities that can only be indirectly manipulated are called *internal quantities*. Examples of these are the speed of the aircraft, the wing's angle of attack and the descent rate. One other kind of quantity is termed a *non-controllable parametric quantity*. These are quantities that, like

Figure 1-1. The Expert's Acceleration Solution

controllable parameters, determine the values of internal quantities, but, unlike controllable parameters, cannot be manipulated by the planner. Air density is an example of a non-controllable parametric quantity in the aircraft domain. The density of the air surrounding the aircraft cannot be manipulated either directly or indirectly, but it significantly alters the aircraft's flying characteristics.

Continuous world changes are more appropriately represented as a graph of simultaneous quantity values rather than as a sequence of discrete world states. We call these graphs *quantity profiles*. They are similar in spirit to event shape diagrams (Borchardt, 1984; Waltz, 1982). An example quantity profile for a simplified automobile domain is shown in Figure 1. It shows a human expert's manipulation, over an interval, of the gas and clutch pedals of an automobile along with the resulting engine revolutions (RPM) and automobile speed.

In our approach, the planning problem is as follows: given an initial state of the world (initial values of quantities), goal values for some *internal* quantities, find a consistent profile of changes of the *controllable parameters* which when combined with the values of the non-controllable parameters bring about the desired changes for internal quantities. The aircraft example above falls into this mold: given the initial positions of the aircraft controls, the heading, the altitude, etc. of the aircraft, find a manipulation of the controls that (for the current air temperature, air density, wind conditions, etc.) reduces the speed by 10 knots while preserving the current angle of descent.

Briefly, the learning algorithm involves observing an expert who solves a problem currently beyond the system's capabilities, constructing a plausible qualitative explanation for why the expert's actions result in

the desired effect, generalizing the explanation in standard EBL fashion (Mitchell, 1986; Mooney, 1986), and fitting observed quantitative points to the resulting general qualitative concept. Concepts constructed in this way can support very efficient planning for the achievement of similar future goals.

2.2 Plausible Qualitative Explanations

A plausible qualitative explanation for observed changes of an internal quantity is an account of how those changes result from increases and decreases in parameter values. It is a “proof” in the system’s plausible qualitative domain theory.

The form of the domain theory is loosely based on Forbus’ Qualitative Process Theory (Forbus, 1984). We use the qualitative predicates INCREASING, DECREASING, and CONSTANT to denote a relation between a quantity and a time interval. They are true in just those cases in which the value of the quantity is monotonically increasing, monotonically decreasing, or constant, respectively, over the entire time interval. The qualitative predicates GREATER-THAN, LESS-THAN, and EQUAL are relations between two quantities and a time interval. They are true if the value of the first quantity has the mathematical relation $>$, $<$, $=$, respectively, to the value of the second quantity over the entire specified interval. Finally, and most interestingly are the two qualitative proportionality predicates Q+ and Q-. Their names and inspiration come from Qualitative Process Theory, but their semantics are somewhat different here. In Forbus’ framework, Q+ and Q- have meaning only after the world of qualitative relations has been closed by assumption. No inference is sanctioned from isolated qualitative proportionalities. Only after knowing all of the world’s qualitative proportionalities can one draw any inferences. We are making no such assumption. (Q+ $q1$ $q2$ i) represents that one quantity, $q1$ is positively qualitatively proportional to another quantity $q2$ over a time interval i . It is to be treated as the conjunction of:

$$\begin{aligned} &(\text{INCREASING } q2 \ i) \Rightarrow (\text{INCREASING } q1 \ i) \text{ and} \\ &(\text{DECREASING } q2 \ i) \Rightarrow (\text{DECREASING } q1 \ i) \end{aligned}$$

while (Q- $q1$ $q2$ i) is short for

$$\begin{aligned} &(\text{INCREASING } q2 \ i) \Rightarrow (\text{DECREASING } q1 \ i) \text{ and} \\ &(\text{DECREASING } q2 \ i) \Rightarrow (\text{INCREASING } q1 \ i) \end{aligned}$$

The symbol “ $\Rightarrow$ ” denotes *plausible* (not logical) implication. Its new semantics raises many issues that are only beginning to become clear. For now view the consequent of a plausible implication as not logically entailed, but only plausibly supported. Domain knowledge is coded in the form of *processes*. Each process has preconditions and a body. The body specifies a set of qualitative proportionalities among quantities. An explanation for an internal quantity change is made up qualitative proportionalities from the active processes and observed qualitative behavior of *parameters*. There may be many different explanations possible for an observed behavior.

Drawing an inference involves propagating a qualitative description through a qualitative proportionality to conclude some new qualitative behavior (e.g., (*INCREASING* $q2$ i) can be propagated through (Q - $q1$ $q2$ i) to conclude (*DECREASING* $q1$ i)). Inferences may be drawn *only* to explain observations. A plausible explanation for an observation is composed of three parts. First, the goal to be explained is a qualitative description of the behavior of some internal quantity of interest. The second component is the observed qualitative behavior of some controllable quantities. These quantities are conjectured to have caused the observed behavior of the goal. The third component is a set of qualitative proportionalities which explain the observed goal behavior in terms of the observed behavior of the controllable quantities.

Explanations are generated in a most-plausible-first ordering. In the current implementation, all qualitative proportionalities are considered equally plausible, so “most-plausible” becomes equivalent to the “simplest”. Given an observation of an expert’s planning, the system constructs the simplest explanation that accounts for the desired qualitative behavior. This explanation is generalized and used by the system to plan. If the explanation is incorrect or incomplete, a later plan based on the explanation may fail. At that time the explanation is rejected in favor of the next-most-plausible one that is consistent with the old and new observations. This explanation is generalized into a replacement planning concept.

This refinement-on-demand of the planning concepts means that the system cannot guarantee that its plan solves a new problem it is given. Any proposed solution may fail, signaling the need for further refinement. Indeed, there would seem to be the possibility that refinement will continue indefinitely with the planner never approaching an acceptable

level of competence. It can be shown that for a fixed problem distribution, the learning algorithm converges to a set of adequate (though not necessarily causally correct) planning concepts.

2.3 Quantifying the Concept

The planner must, of course, produce a *quantitative* solution. It is not sufficient to know that the throttle must be closed; the planner must decide by how much. Thus, a purely qualitative representation is not convenient for planning. (However, see (Hogge, 1987) for an interesting but computational intensive approach.) A planner must produce a solution of sufficient quantitative precision to execute in the world, and this generally requires a quantitative planning concept. It is obtained by calibrating the generalized *qualitative* explanation with the *quantitative* value of actual observed world points so they can be retrieved when needed. The concept describes the behavior of the world at non-observed points simply by interpolating between known values.

The explanation which, recall, is purely qualitative, provides two important constraints on the final quantitative concept. It A) identifies the parameters that are relevant to achieving the specified goal and B) assures a strong constraint on how the internal quantity value may change with parameter changes: the internal quantity's function must be monotonic in each of the parameters identified by the explanation. The actual monotonicity relation is computed through transitivity of qualitative proportionalities (Q+ and Q-) in the explanation. The quantitative concept is a numeric function which is constructed to approximate the actual internal quantity function. Its arguments are the parameters identified in the explanation. Initially, there is no *quantitative* information available to constrain the function but *qualitatively* its behavior is known: it must obey the explanation's monotonicity constraints.

The numeric function is calibrated by asserting the sequence of observed tuples. Each tuple is composed of an observed quantitative value for the internal quantity together with the corresponding observed quantitative values for the parameters. There is one such tuple for each time step in the observed expert's solution. Thus, the numeric function acts as a repository for the observed quantitative behavior. Values for the internal quantity at non-observed points are approximated by interpolating between the known points.

The resulting quantitative concept allows planning and execution to be interleaved. At each step changes are selected for the controllable parameters that, according to the numeric approximation function, most effectively reduce the difference between the current and desired values for the goal quantities. Once the manipulations are performed, the actual values observed from the world are compared to those predicted by the planning concept. If the predicted change in the goal quantity is within a pre-specified error threshold, another planning/execution step is performed. Any difference beyond the error bound causes the concept's approximation function to be refined by asserting the newly observed point to participate in interpolations. If the refinement is inconsistent with the concept's qualitative monotonicity constraints, the original explanation must have been in error so the explanation is rejected and the next-most-plausible explanation, consistent with the new observation, is constructed and generalized.

Planning, apart from the execution, consists of the decisions about how to manipulate the controllable parameters. This includes any quantitative or qualitative concept refinement. Planning time is the sum of the CPU time expended in planning decisions over all time steps.

3. An Example of Continuous Domain Planning

A system, called MARIO, has been implemented and tested on several domains. In one of the domains, the system plans to achieve different speeds in a simplified single-gear manual transmission massless automobile. The goal is to produce control motions that result in as smooth a change as possible from the initial speed to the final speed. In fact, the "real world" automobile is a numerical model whose equations directly incorporate the values of controllable parametric quantities (e.g., the gas, clutch, brake, and window position).

GAS, CLUTCH, and BRAKE are symbols in the domain knowledge that represent the position of the gas, clutch, and brake pedals, respectively. These are controllable parameters. For each, there is a maximum rate at which it can be changed. In the numerical automobile simulator, increasing the GAS and BRAKE correspond to depressing an actual automobile pedal. Increasing the CLUTCH corresponds to letting up on an actual automobile pedal. ENGAGEMENT is the percentage of power transmitted to the wheels through the clutch. RPM is the rota-

tional speed of the engine. SPEED is the speed of the car. GRADE is the hill gradient, a non-controllable environmentally-set parameter. TEMP is the engine temperature.

Figure 2 shows MARIO's background knowledge for this domain. It consists of two qualitative processes. One, **ENGINE-RUNNING**, specifies relations among quantities that may plausibly be believed whenever the engine is on. For example, the first line states that as the gas pedal is depressed, the gas flow to the engine may increase and vice versa. The second process **CAR-MOVING**, contains descriptions that may be believed whenever the car is moving forward.

The processes given in Figure 2 are not the only ones that might have been used to describe the workings of our automobile. One of the strengths of this approach is that it is relatively insensitive to how the domain knowledge is crafted. One might write a simpler or a more complicated set of processes (specifying relations among the carburetor butterfly valve position, venturi flow, etc.). Many alternative domain theories would work as well for our chosen task.

In the first problem of the sequence, the system is given the goal of achieving a speed of 64 MPH from a dead stop with the engine on and idling. The problem also specifies initial and final engine RPMs. For simplicity, we will focus on the achieving the SPEED parameter, however, MARIO performs analogous reasoning in each problem to achieve the desired RPM value. Initially, MARIO has no planning concepts and thus cannot solve the problem unaided. It asks for an expert's solution. Using keyboard/mouse input the expert manipulates the controls to achieve the goal. Some of his manipulations and their relevant results are shown in Figure 1.

In this example there is but a single explanation interval, so the time interval arguments are omitted for simplicity.

The first plausible explanation consistent with the expert's solution is shown in Figure 3A. The explanation is the shortest inference path connecting the observed acceleration to observed control manipulations. Two inferences are employed, each using one qualitative proportionality from the domain theory. A simple breadth-first search would suffice to realize the simplest-first explanation production. In practice, inefficiency of such straightforward searches becomes intolerable for more complex explanations. MARIO generates explanations using a modified chart parser which attempts to "parse" the goal behavior in terms of

Figure 1-2. Domain Processes

observed controllable behavior with qualitative proportionalities from the active processes serving as the grammar re-write rules.

This explanation conjectures that the speed is increasing because the speed is qualitatively positively proportional to the clutch engagement which is qualitatively positively proportional to the clutch pedal position which is observed to be increasing.

Thus, the speed is conjectured to be a non-decreasing continuous function of the clutch position alone. If the analysis were correct and MARIO knew this function, then any desired speed profile could be achieved by manipulating the clutch accordingly. With the observed expert behavior and the qualitative explanation, MARIO has sufficient information to construct an approximation to the hypothesized real-world function.

The functional relation of the speed to the clutch can be captured as a two-dimensional curve whose ordinate is the car's speed and whose abscissa is the clutch position. The qualitatively positively proportional constraint requires the slope of the curve to be non-negative. This constraint is derived from the explanation. In addition, there quantitative constraints: the curve, whatever it is, must pass through each of the clutch-speed pairs observed during the expert's solution. There are 89 discrete observations during the interval yielding 89 clutch-gas pairs.

Figure 1-3. MARIO's First and Second Explanations of the Expert's Behavior

Figure 1–4. Planning with the Clutch Alone

MARIO employs piecewise linear interpolation. We might elect to plot all 89 points, connecting points with adjacent clutch values with straight lines. As it happens, however, 8 points can be selected so that the resulting piecewise linear curve lies very close (within the allowed noise threshold) to the remaining 81. The resulting piece-wise linear approximation function is shown in Figure 4A. As the clutch is increased, the speed increases in response.

A second acceleration problem is given to MARIO. This time, the car is to accelerate from 0 to 20 MPH. Since MARIO now has a relevant planning concept it does not ask for an expert's solution but attempts to solve the problem itself. Mario selects a clutch profile which according to the interpolation curve of Figure 4A will yield a constant acceleration to 20 MPH. It begins feeding the progression of clutch values to the numerical simulator, monitoring the resulting speed values. Almost immediately a discrepancy is found between the observed and anticipated speed values. The discrepancy is not only larger than the allowed noise threshold, it violates the monotonicity constraint of the explanation. Figure 4B shows the interpolation curve with the discrepancy point asserted. The discrepancy point is just to the left of and significantly *above* the second original interpolation point. These two points are circled. To pass through both observed points, the speed must *decrease* by more than the noise threshold as the clutch position *increases*. The

Figure 1–5. The Third Explanation

contradiction means that the original plausible explanation is not, in fact, adequate. The explanation and the planning concept based upon it are rejected.

The system constructs the next-most-plausible explanation for the original data augmented by the new point that resulted in the failure. This explanation is shown in Figure 3B. This explanation conjectures that the SPEED can be adequately controlled using the gas pedal alone. MARIO is given a third problem in the sequence. In attempting to apply the new concept the system also meets with nearly immediate failure. The next-simplest explanation is constructed. It is more adequate than either of the explanations of Figure 3 and is shown in Figure 5.

This explanation indicates that the SPEED is a function of both the GAS and CLUTCH controllable parameters. The system finds that 35 of the 91 observed points are sufficient. With these, the values of the remaining points can be predicted accurately by interpolation. Notice that the explanation is not, in fact, correct. For example, it misses the effect of GRADE upon SPEED. There is an implicit assumption that GRADE can be disregarded when solving acceleration/deceleration problems. This is valid only in certain planning contexts as in the Champaign/Urbana area, where the only two hills are man-made for children's sledding pleasure in the winter. If a later planning problem depends on the missing GRADE or BRAKE parameters, the above concept would be eliminated in favor of one more faithful to reality. However, this concept is adequate for all problems in the problem distribution. A fourth problem of the sequence is generated. The system generates the

Figure 1-6. The Planner's First Adequate Solution

adequate solution shown in Figure 6.

Notice that the speed is not smoothly increasing. The controllable parameters' values in Figure 6, particularly the CLUTCH, exhibit a fair amount of "hunting" for the right values. This is due to interpolation error; the interpolation surface predicts one value for the internal quantity SPEED while the numerical automobile simulator in fact yields another. However, the discrepancies are not so great as to violate the qualitative constraints. Such discrepancies mean only that the assumed linear relationship in the region is inadequate. For each such discrepancy, the newly observed point is asserted to the approximation surface. This breaks the original region into a number of smaller regions which more accurately approximate the true surface. Once again the surface agrees with known reality at least to the tolerance of the noise threshold. As it happens, 27 additional points are asserted to the interpolation function during this first successful solution. The function is then approximated by a total of 42 points. After 99 additional acceleration and deceleration problems the interpolation function contains 134 points, mostly added during early problems. The system produces the solution shown in Figure 7. This problem is a deceleration problem. Once an adequate functional relation has been discovered for the SPEED and RPM quantities, the planning concept is applicable to any achievable goal profile, including decelerations.

As can be seen, the SPEED and RPM's are decreased smoothly and linearly. The system's manipulation of the control parameters are also smoother. Smoothness of control manipulations is not an explicit goal

Figure 1–7. A Later Solution

of the system. This property emerges from the basic smoothness of the world (in this case the numeric simulator) and the smoothness of the desired SPEED and RPM goal transitions. However, notice the non-linear manipulation of the controls. This is due to a non-linearity within the finite difference simulator. The flow of gas is a non-linear function of the gas control position. The learning system is not told of this non-linear relation. Rather it is discovered through experience. Due to the particular operating characteristics, the non-linearity is manifested both in the CLUTCH manipulation and in the manipulation of the GAS parameter.

4. Empirical Evaluation

Figure 8 shows a typical progression of planning behavior following the initial observation of the expert. In the experiment MARIO is given 10 randomly generated acceleration and deceleration problems drawn from a fixed distribution. In each of the 10 problems MARIO is asked to achieve a desired SPEED and RPM combination from a given initial automobile configuration. The 10 test problems are preceded by a single training instance in which a human expert presents his solution to a problem drawn from the same distribution. It is important to note that MARIO continues to learn during execution of its test problems. No additional expert intervention or solutions are provided, but MARIO can observe the quality of its solutions as they are executed on the

Figure 1–8. Empirical Performance

numerical simulation automobile.

Each test problem requires on-going manipulation of the controls over an interval of time. At each time step the controls can be moved in either direction subject only to their individual limit on maximum rate of change. At each time step a planning decision must be made as to what fraction of the maximum change to manipulate each control. Each problem requires between 20 and 100 time steps each to solve. More time steps are needed when the goal configuration of the car turns out to be further from its initial configuration.

The graphs show cumulative average information. The heavier line plots the CPU time for a time step in seconds averaged over all of the problems thus far. The lighter line plots the average cumulated error in miles per hour for all planned points thus far.

MARIO's convergence to competent problem-solving appears to be fast in practice for the few domains we have examined. Both accuracy and CPU run time improve significantly over the 10 trials. The expected error between anticipated and sensed automobile SPEED falls to a small fraction of one mile per hour. This is well below the failure threshold. Very little concept refinement is performed after the seventh problem.

In addition to improved accuracy, MARIO's planning time (the CPU time expended in choosing the GAS and CLUTCH manipulations) decreases with experience. The reason for faster planning is that the cost of adding a new point to the interpolation function is high compared to

the cost of employing the function without refinement. As experience with the concept increases, fewer new interpolation points are added.

The shape of the curves in Figure 8 are quite typical for other distributions and other domains. A period of rather intense concept refinement gives way to an asymptotic limit for both accuracy and CPU time. The number of problems required to approach the asymptote depends on the topology of the physical system's state space, the range of the distribution of problems, and the shape of the distribution. In this domain, the automobile's state space is non-linear but smooth. The problems were chosen from a uniform distribution of a relatively narrow section of the state space - roughly over 25 the SPEED and the RPMs.

5. Influents

Next we wish to question why MARIO seems to work so well. There are a number of candidate sources of power that may be underpinning its successful behavior. One possibility is the robustness of numerical interpolation: continuous domain planning requires manipulation of numerical parameters and interpolation is an old and rather well-understood field. Another possible reason for success might be that the continuous domains examined lend themselves well to the particular combination of qualitative and quantitative reasoning that MARIO performs.

We believe that the true reason for success is deeper. We maintain that MARIO is successful primarily because it employs a more flexible kind of explanation-based learning in which generalized explanations are more interesting than logical proofs. The key features are that explanations are conjectured rather than proven, and that generalized explanations can be calibrated with information gained empirically from the world. This dual procedure of conjecturing and calibrating can be abstracted away from the continuous domains of MARIO. Indeed, the technique is more general than planning and can be used to define a flexible plausible inference procedure which we call *Influent Logic*.

To trace and isolate its source of power, we first observe that MARIO clearly possesses the non-brittleness property outlined in section 1: As new observations of the world are made, (different profiles of an automobile's speed) they are integrated into the system's internal model. If the observations are consonant with the current planning concepts describing the phenomenon everything is fine. If, however, a world ob-

servation contradicts some prediction made by a planning concept, the concept is refined to bring it in line once again with the observed world.

This behavior is due to MARIO's forms of concept refinement. If the quantitative prediction is out of bounds while respecting the underlying qualitative explanation, then simply adding additional interpolation points suffices as a refinement. If, on the other hand, the observation is so far off so as to violate the qualitative constraints derived from the underlying plausible explanation, then the explanation itself is discarded in favor of the next-most-plausible one. The true source of power is derived from the combination of *analytically*-driven hypotheses which are *empirically* calibrated, confirmed, or rejected. The qualitative/quantitative difference is merely one method of supporting the conjecture/calibrate cycle.

Unlike conventional inference models, in this scheme deriving an expression from the system's background beliefs is insufficient for the system to *believe* the expression. The fact that MARIO can construct a connection between an observed change in SPEED and an observed change in the opening of the driver's window does not require the world to behave accordingly.

The set of *logical* theorems of the background knowledge representations forms only the *hypothesis* space of plausible theorems. Further empirical confirmation is required before any hypothesis is elevated to the status of a *theorem*.

We now show how the important idea of plausible reasoning can be applied to more conventional symbolic representations, and, in so doing, outline the beginnings of a theory of influent logic.

5.1 Plausible Representations

As with a conventional inference system, we begin with a set of expressions Δ given to the system as background knowledge. Additionally, we require a generator of observations which provides an unlimited sequence of independent observations of the world drawn according to some fixed but possibly unknown distribution. The expressions of Δ are syntactically similar to a set of implications. -Semantically they are quite different. Each expression can be viewed as a conventional deductive implication but with missing antecedents. We refer to the missing antecedents collectively as the *implicit context* of the expres-

Figure 1–9. The Semantics of an Influent

sion. The implicit context is the set of additional constraints sufficient to guarantee that the plausible implication deductively holds. We refer to these plausible expressions as *influent*s to emphasize their difference from material implication. Figure 9 shows the relationship between an influent and a deductive implication. Asserting an influent specifies that its antecedents are relevant to determining its consequent in some contexts¹. An influent also specifies the direction of influence between the antecedents and consequent—they may be positively or negatively related.

5.2 Inference Over Influent

To build plausible explanations from a set of influents we need an inference mechanism. Influent may be combined with other influents in the same way deductive implications are chained together. Unifying a antecedent of one rule with the consequent of another provides the first influent inference mechanism. One additional inference mechanism is not conventionally employed: the join, which allows combining several influents with the same consequent (or its negation). A joined set gives no information regarding the applicability of its component influent chains, but it does guarantee that there is some combination of these chains which correctly predicts the conclusion of the influent set over the relevant problem distribution.

1. The influent determines its consequent as in relevance logics while allowing the axiom set to remain open as in intuitionistic logics.

Figure 1–10. Construction of Influent Sets

Plausible explanations are generated by repeated application of these inference mechanisms. Such explanations are generated only in response to observations and they must be consistent with all of the observations they explain. Each plausible explanation defines an *influent set* (see Figure 10). An influent set specifies that the truth values of its leaf expressions determine the truth value of its consequent, without completely specifying the mapping. As such it can be viewed as a *conjectured* determination (Grosz, 1989) embellished with the directionalities. Both serve as a specification of the set of determining factors for some value in the domain. Both are under-constrained and are consistent with any number of initial observations. As examples are encountered, the initially under-constrained mapping between antecedent and consequent truth values becomes better known. This is the calibration concept refinement of the plausible system.

Unlike Russell's determinations, however, an influent set may be contradicted by a later observation. There is no guarantee that a plausible explanation actually *does* determine its consequent due to the implicit contexts of the underlying constituent influents. When such a contradiction arises, the explanation that gives rise to the influent set is rejected in favor of the next-most-plausible explanation. Another influent set is constructed and the calibration process repeats. This completes the conjecture/calibrate cycle that underlies our plausible reasoning system.

5.3 Influent Sets

For every derivation consistent with an observation there is a corresponding influent set. The plausibility values of the influents which participated in the derivation can be used to provide a partial ordering on the *a priori* plausibility of the explanation and, therefore, the influent set. This allows a most-plausible-first ordering strategy on the set of possible conjectures. The conjectured concepts are then tuned by observation. We must also be able to know when a conjectured concept must be rejected. Influent sets must possess the following three properties with respect to the known observations. Violation of any of the three means the influent set is not an adequate concept for describe the observed phenomenon. The three conditions state that an influent set must be *deterministic*, *predictive*, and *relevant* over the set of observations:

This means that the consequent must be completely determined by the preconditions of the influent set over the set of observations. There cannot be two observations which agree on the truth value of all of the preconditions but do not agree on the truth value of the consequent.

This is the same constraint placed on Russell's determinations.

The second consistency property is predictiveness. It states that when an influent is needed to distinguish between two observations, the influence it has on the consequent must be in the predicted direction. If two observations differ only in the truth values of the preconditions of several positive influents then the consequent must be directly related to the preconditions (it cannot be true only when the preconditions are false). This constraint ensures that positive values assigned to the preconditions can only have a positive effect on the consequent. The analogous negative relationship must also hold for negative influents.

The final consistency property is relevance. An influent set is relevant over a set of observations if the preconditions of each influent are necessary for uniquely determining the value of the consequent. No influent chain that superfluous to the prediction of the observations is allowed in the influent set; each influent chain must be the determining factor in some circumstance otherwise it could be eliminated from

the explanation with no effect on the learned function. Relevance is required since it disallows plausible theorems which are correct in the sense that their preconditions determine their consequent, but are useless since they have so many irrelevant influents. Any explanation that satisfies these three properties determines a function which can be used to predict the domain. Which is necessary for a robust system. Thus the PI-EBL approach can be characterized as a search for plausible theorems—explanations which are empirically shown to be deterministic, predictive, and relevant.

5.4 PI-EBL

Plausible inference and explanation-based learning together form a framework for reasoning which is less brittle than traditional EBL. Unlike EBG and EGGS, the PI-EBL approach admits multiple incompatible explanations of the same goal concept making the approach similar in spirit to (Cohen, 1990; Eskey, 1990). However, we crucially introduce the calibration phase. The PI-EBL algorithm is as follows:

One criticism of traditional EBL is its limited use of training examples. PI-EBL makes much more comprehensive use of its examples. They are simultaneously used to fill three roles in the explanation pro-

Figure 1-11. A Plausible Domain Theory

cess. As with traditional EBL the example provides the fact to be explained by the plausible inferencing mechanism. Examples are also used to verify or reject plausible explanations generated by the inferencing mechanism. This is done in a generate and test fashion. If an observation presented to the system causes failure of the current hypothesis to be deterministic, predictive, or relevant over the set of observed examples, then the hypothesis is rejected and the inferencer is used to generate another plausible explanation. As plausible explanations are being generated and monitored for correctness the observed examples are used for a third purpose. Associated with every influent set in a plausible explanation there is a function mapping the the preconditions of the influent set to its consequent. Observations are used to constrain these functions.

5.5 An Example of the PI-EBL Algorithm

We present a simple example of the PI-EBL learning algorithm to supplement its description and as a concrete example of its advantages over traditional EBL.

Figure 11 shows a plausible domain theory which is used to understand the effects of the “Left” operator, which pushes the block specified by the operator left one unit. For simplicity of description we will use a one dimensional blocks world with unit sized blocks. Notice that in Figure 11 all of the influents have no functions as arguments to their consequent predicate. This is done by convention. Joins are defined for trivial (renaming) unifications only. This causes no problem since any functions in the consequent can be moved to the antecedent using an additional “Eq” (Equal) predicate. The first influent in the plausible

Figure 1–12. Observations Used To Refine “Left” operator

theory shown in Figure 11 describes the action of the “Left” operator (moving the block’s position). The remaining influents provide general knowledge about the blocks world. The influents in the plausible theory can be roughly translated in to english as:

“When pushed left blocks move left.”

“Blocks typically don’t move.”

“A block cannot occupy two different positions at the same time.”

“Two different blocks are typically not in the same place at the same time.”

Individually these rules seem reasonable but incomplete (they have exceptions). As a group, however, they make inconsistent predictions about the effects of the “Left” operator.

The PI-EBL algorithm learns how these influents interact in the world being modeled through observations from that world. Figure 12 has a set of observations which we will use to demonstrate how the PI-EBL algorithm converges on a predictive set of explanations for the domain it models. The first observation is of the red block which does not move. This is explained by the trivial explanation containing just one influent: $At(block, x, state) \wedge Eq(newstate, Do(op, state)) \succ At(block, x, newstate)$.

The second observation is not consistent with this no movement prediction. And so an explanation is sought which explains both observations. Figure 13 show such an explanation. It overrides the “blocks typically don’t move” with “Move left reduces the position of block by

Figure 1-13. Explanation For Observations 1 And 2

one” to compute the position of the block. Together these predict the position of the block in the first two observations. Determining that the block has moved, however, is not logically equivalent to concluding that it is no longer in its previous location. The third component of the join is used to “override” the conclusion that the block remains in the same position. It roughly says, if the block has moved as a result of the “Left” operator it cannot continue to occupy the original location. The join function shown in the figure shows what truth value is assigned to the join’s consequent as a boolean function of its constituent explanations. These values are assigned by the PI-EBL algorithm such that the resulting join is deterministic, predictive, and relevant over the observations the system has been given. In this case the table is completely determined by the two observations. The three “*” in the table will never be filled in by the PI-EBL algorithm since no examples can be observed in these categories, those combinations of truth values are not consistent with any possible observation. Either they are internally inconsistent or they assume an impossible situations, such as two blocks in the same position.

The third observation is not consistent with the explanation in figure 13. The explanation predicts that the block should move, but as we observe the block remains in place. Notice, if the block were to move

Figure 1-14. The Final Plausible Explanation

as the rule suggests it would be in the same position as another block. The last rule in our domain theory allows us to explain this apparent contradiction. It is used to generate another exception to the move left operator. This exception is generated from chaining the first and fourth influents, and is shown as the last component in Figure 14. Again the join mapping is completely specified by the observations made. This plausible explanation completely determines the new value of the At predicate after an operation. It predicts that a block will remain in place unless the move left operation is applied to that block and there is space available to the left of the block.

PI-EBL was able to quickly learn the correct effects of a very incompletely specified operator. Its rapid convergence was due to its guidance by a simple theory about the blocks world. The simplicity of the specification of the move left operator as well as the simplicity of the plausible theory is a result of the flexibility of the plausible representation. The implicit context allows plausible theories to admit imprecise knowledge like “blocks typically don’t move.” Such knowledge is much easier to obtain than a complete and correct theory, and can still provide strong guidance for understanding the domain.

6. Conclusions

Reasoning about continuous attributes is an important but much neglected facet of planning. In day-to-day life most human planning involves manipulation of continuous parameters of the world. We describe an EBL approach in which *plausible* qualitative explanations are conjectured as interpretations of real-world observations. When generalized, these explanations form the qualitative framework on which a quantitative concept can be constructed. The quantitative component is formed by numeric calibration within the general qualitative constraints gleaned from the explanation. Calibration is empirical, using observed real-world measurements.

The approach was in part motivated by the need for efficient planning so that it might be extended to the kinds of flexible real-time control that people perform in their daily lives. A system, called MARIO, has been implemented and tested. While we are still exploring the system's characteristics, evidence we present indicates that the approach is capable of both accurate and efficient reasoning about continuous real-world quantities.

Another design motivation of MARIO was to better capture our intuitions concerning explanation-based learning. Confining explanations to be logical proofs relegates EBL systems to sterile micro-worlds in which training examples seem superfluous and knowledge-level learning is not possible.

MARIO's plausible explanations support a richer notion of EBL that much better captures the original excitement we felt about the approach. Thus, we sought to analyze the reasons for MARIO's success. Somewhat to our surprise, the attributes of MARIO we found so attractive had little to do with continuous quantities or qualitative representations. An important feature of MARIO's inference ability is that the world is left open; it abandons the law of excluded middle. In addition, the notions of "proof" and "derivation", so intimately intertwined in standard logic, are separated. In itself, a sound derivation of an expression from correct prior beliefs no longer constitutes a proof of the expression's truth or accuracy. However, such derivations (or explanations) can be interpreted as plausible conjectures which may become believed if empirically validated. MARIO's power is derived from the fact that its inferential process intimately depends upon observing the world. Observations lead to conjectured interpretations which, after

generalization, are calibrated and tested against further observations.

A major strength is the ability to acquire the simplest adequate reasoning concept. Thus, a multitude of irrelevant features of the world can be implicitly ignored. MARIO, for example, never entertains the effect of hill gradient on automobile speed. Indeed, an ability to adjust controls for driving up and down hills is of little use in Urbana. In San Francisco, however, this skill may be crucial. If the system were employed in a San Francisco-like terrain, the need to adapt to failures would soon lead to the incorporation of hill-gradient influents. A key to efficient reasoning about the real world must be to avoid ever considering most such influences. If a planner must enumerate them all, if only to post assumptions that they are irrelevant, it will be paralyzed by the complexities of the real world.

MARIO's reasoning in the face of real-world subtleties is formalized into *influent* logic. It holds promise for reasoning in situations where there is unavoidable complexity and no a priori micro-world idealization. It can be seen as an approach to overcoming the qualification problem. The problem is particularly pernicious in planning applications. Plausible inference with EBL provides an alternative although at a cost. It inextricably binds learning with reasoning and relies heavily upon observable behavior of the real world. The reasoner refines its reasoning abilities when the reasoner's conclusions are contradicted by world observations. Contradictions are one of two forms. Calibration refinements result in further constraints on existing concepts. Interpretation refinements involve reanalyzing observations to result in different conjectured interpretations. Since failures are necessary, the approach precludes any logical proof that the reasoner's answer is correct, although confidence in each reasoning concept increases with experience.

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